

AI-Based Face Extraction and Clustering Pipeline

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EXECUTIVE SUMMARY

This document describes a **production-ready, four-workflow face recognition system** designed to handle real-time ingestion, intelligent clustering, persistent storage, and multi-use-case retrieval across enterprise and event photography domains.

Key Capabilities:

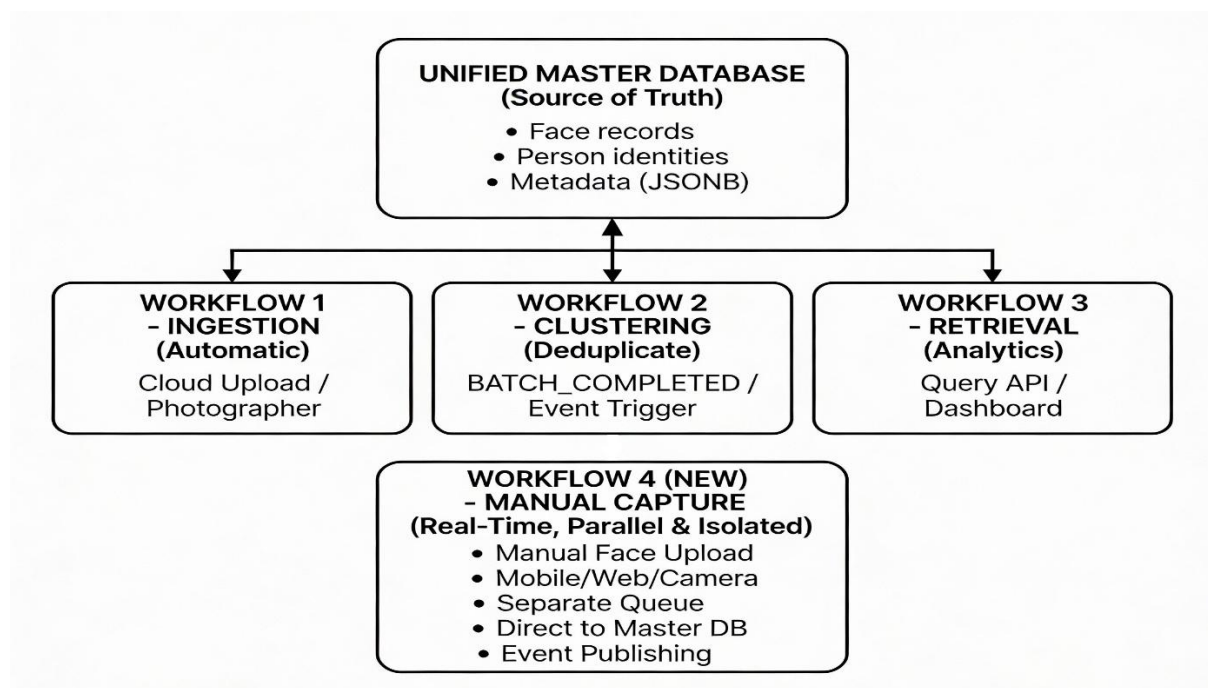
- Automatic batch ingestion of images with real-time embedding extraction
- Intelligent DBSCAN clustering for unique person identification
- Real-time analytics and interactive dashboards
- **NEW:** Real-time manual face capture with use-case-driven routing
- Complete isolation between workflows for maximum reliability

Core Technology:

- InsightFace-REST for face detection and 512-D embedding extraction
- FAISS (Facebook AI Similarity Search) for vector indexing
- PostgreSQL for master data management
- Google Cloud Platform (Pub/Sub, Storage, Compute, Database)
- Kubernetes for container orchestration and auto-scaling

SYSTEM ARCHITECTURE OVERVIEW

The system consists of **four independent, asynchronous workflows** that operate in parallel without blocking each other:



WORKFLOW 1: INGESTION (BATCH PROCESSING)

Purpose

Automatically process high-volume image uploads from photographer devices with minimal latency.

Input

- Photographer devices upload photos to Google Cloud Storage
- Event triggers on object creation

Process

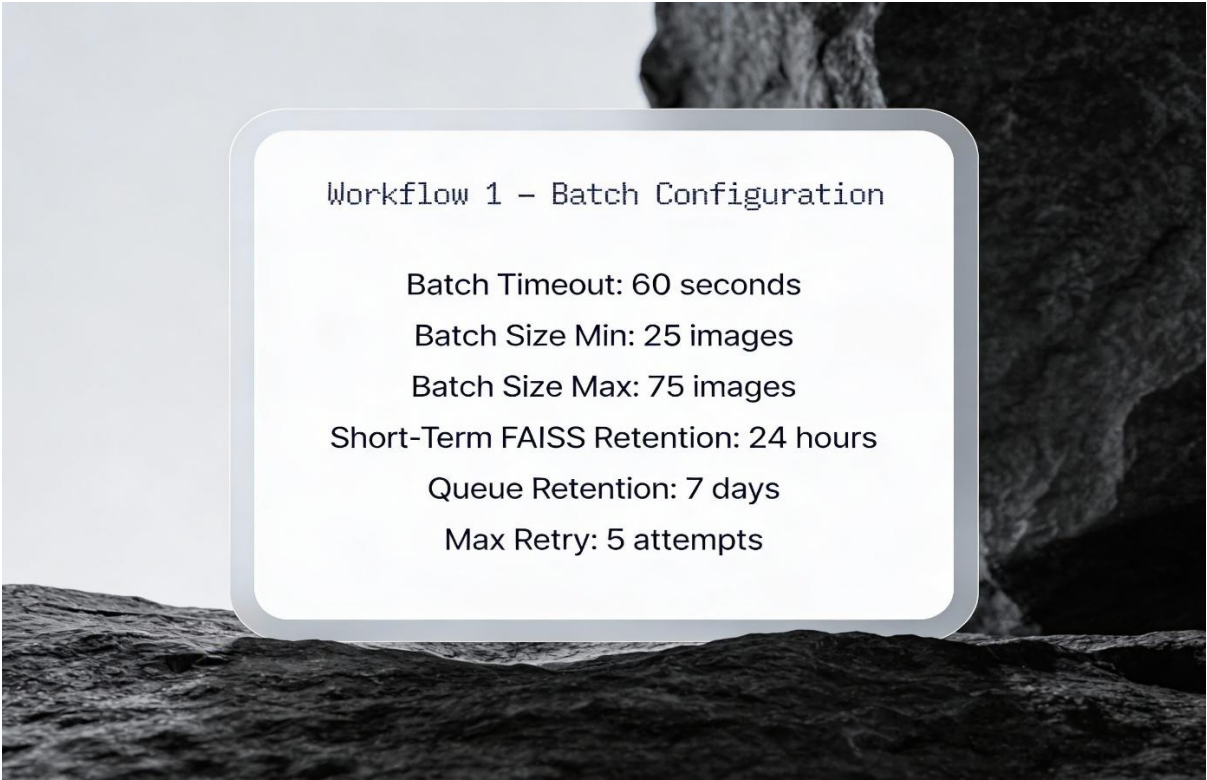
1. **Batch Grouping:** Group 25-75 photos together
2. **Message Publishing:** Send batch to wfl-image-batch-queue (Pub/Sub)
3. **Auto-Scaling:** Spin up embedding workers based on queue depth
4. **Face Detection:** Load batch, detect faces per image
5. **Embedding Extraction:** Extract 512-dimensional vectors using InsightFace R50
6. **Demographics Classification:** Classify gender ('M'/'F'/'U') and age group ('20s', '30s', etc.)
7. **Short-Term Storage:** Write embeddings to in-memory FAISS index
8. **Metadata Store:** Store face metadata in Redis/Firestore
9. **Event Publishing:** Emit BATCH_COMPLETED event

Output

- **Short-term FAISS DB:** Indexed embeddings for current session
- **In-Memory Metadata Store:** Detection confidence, demographics, image paths
- **Event Stream:** Ready for Workflow 2 processing

Key Metrics

Metric	Value
Latency (p95)	500ms - 2s per image (amortized in batch)
Throughput	50-200 images/minute
Batch Size	25-75 images
GPU Requirement	1x A100 per worker
Error Rate	<1%

A presentation slide with a dark, rocky background. A white rounded rectangle in the center contains the text for Workflow 1 configuration.

Workflow 1 – Batch Configuration

Batch Timeout: 60 seconds

Batch Size Min: 25 images

Batch Size Max: 75 images

Short-Term FAISS Retention: 24 hours

Queue Retention: 7 days

Max Retry: 5 attempts

WORKFLOW 2: CLUSTERING & STORAGE

Purpose

Deduplicate face embeddings to identify unique persons and create persistent indexed records.

Input

- BATCH_COMPLETED events from Workflow 1
- Short-term FAISS embeddings + metadata

Process

1. **Load Batch:** Retrieve embeddings from Short-term FAISS DB
2. **Group by Demographics:** Separate by gender/age for better clustering
3. **DBSCAN Clustering:**
 - Algorithm: Density-Based Spatial Clustering
 - Parameters: $\text{eps}=0.6$ (cosine distance), $\text{min_samples}=2$
 - Metric: Cosine similarity (suitable for normalized embeddings)
4. **Cluster Assignment:** Assign each face to a cluster (or mark as noise)
5. **Person ID Creation:** Map cluster centroids to unique person identities
6. **Persistent FAISS Merge:** Add batch embeddings to historical FAISS index
7. **Master Database Update:** Record `person_id`, cluster information, demographics

8. Event Publishing: Emit STORAGE_COMPLETED event

Output

- **Persistent Historical FAISS DB:** All embeddings indexed by person_id (100M+ scale)
- **Master Database:** Person records with unique identities and image metadata
- **Event Stream:** Ready for Workflow 3 consumption

Key Metrics

Metric	Value
Latency (p95)	<5 seconds per 50-image batch
Throughput	1-5 batches/minute
Clustering Quality	Silhouette score: 0.70+
Duplicate Elimination	30-50% typical
CPU Requirement	8 vCPU per worker
Error Rate	<0.5%

DBSCAN Parameters :

eps: 0.6 # Cosine distance threshold (0=identical, 1=opposite)

min_samples: 2 # Minimum cluster size

metric: cosine # Normalized embedding distance

WORKFLOW 3: DATA RETRIEVAL & ANALYTICS

Purpose

Provide real-time dashboards and analytics without impacting ingestion/clustering operations.

Input

- Read-only queries to Master Database
- Real-time dashboard requests

Process

1. **Query Master Database:** Execute analytics queries

2. **Compute Statistics:** Gender/age distributions, person counts, photographer stats
3. **Build Galleries:** Retrieve all images for specific persons
4. **Format Responses:** Package data for API consumption
5. **Serve via REST/GraphQL:** Stream to dashboard frontend

Output

- **Dashboard Data:** Live statistics, histograms, trending metrics
- **API Responses:** <500ms latency for typical queries
- **Person Galleries:** Image thumbnails and metadata per person

Example Queries

Workflow 3 – Data Retrieval SQL Snippets

---Live statistics

```
SELECT gender, COUNT(*) as count From master faces
  WHERE DATE(capture timestoop); CURRENT DATE
  GROUP By gender;
```

-- Top photographers

```
SELECT photographer id, COUNT(Distinct person id) as unique
  FROM master faces
  WHERE DATE(capture timestenaps); CURRENT DATE
  GROUP BY photographer id
  ORDER BY uniquepersons DESC
  limit 10;
```

-- Person gallery

```
SELECT image path, capture timestope, photographer id
  FROM master faces
  WHERE person id = $1
  ORDER BY capture Timestopp DEC;
```

Key Metrics

Metric	Value
Latency (p95)	<500ms
Queries/Second	50-200
Dashboard Uptime	99.9%
Data Staleness	0-5 minutes (acceptable)
Error Rate	<0.1%

WORKFLOW 4: MANUAL FACE CAPTURE (NEW ⭐)

Purpose

Real-time manual face uploads for enterprise use cases: employee tracking, security, visitor management, access control, attendance, and event photography.

Key Differentiator from Workflow 1

Aspect	Workflow 1	Workflow 4
Trigger	Automatic cloud upload	User/system action
Batching	25-75 images grouped	Individual faces
Latency	500ms-2s per image	<1 second per face
Clustering	Via DBSCAN (WF2)	None - direct storage
Metadata	Basic (confidence)	Rich (use_case, employee_id, threat_level)
Integration	Photos only	Multi-system (HR, security, access control)
Queue	wf1-image-batch-queue	wf4-face-capture-queue (isolated)
Database Path	WF1 → WF2 → Master DB	Direct to Master DB + Persistent FAISS

Input

- Manual face uploads (mobile app, web UI, surveillance cameras, access terminals, kiosks)
- Face image bytes + use-case-specific metadata

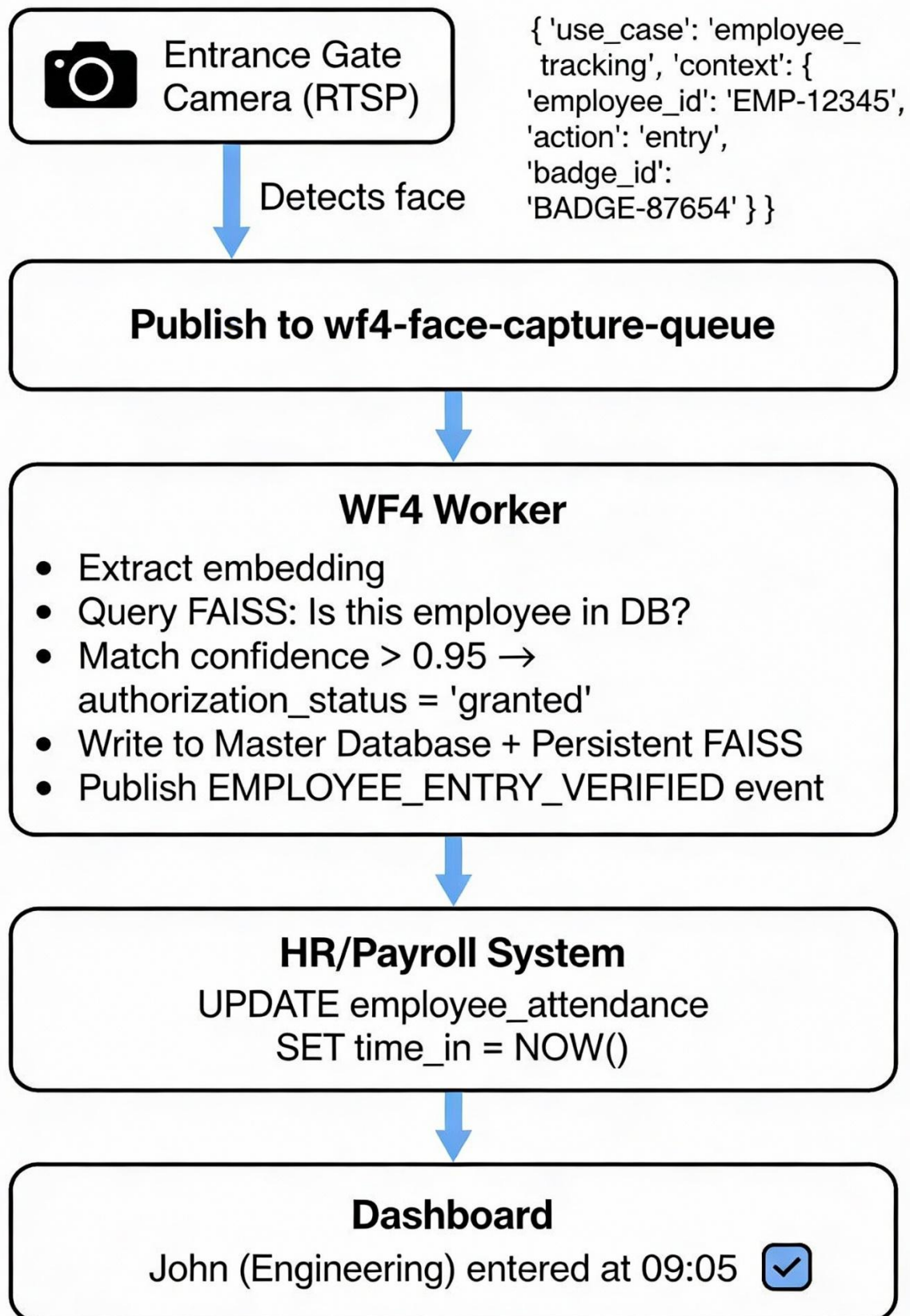
Process

1. **Load Face Image:** Decode from base64 or raw bytes
2. **Extract Embedding:** Generate 512-D vector using InsightFace
3. **Classify Demographics:** Determine gender and age group
4. **Generate Face ID:** Create UUID for this face record
5. **Route by Use Case:** Execute handler based on use_case field
 - **employee_tracking** → Verify against employee DB, publish clock event
 - **visitor_management** → Create temporary badge, notify host
 - **access_control** → Check authorization level, allow/deny
 - **security_monitoring** → Detect unknown persons, escalate threat
 - **attendance_system** → Biometric verification, log time
 - **event_photo** → Tag VIP/guest, organize gallery
6. **FAISS Matching:** Query Persistent FAISS to find similar faces
7. **Write to Databases:**
 - Master Database: Face record with rich metadata
 - Persistent FAISS: Embedding immediately searchable
8. **Publish Events:** Send use-case-specific notifications to external systems

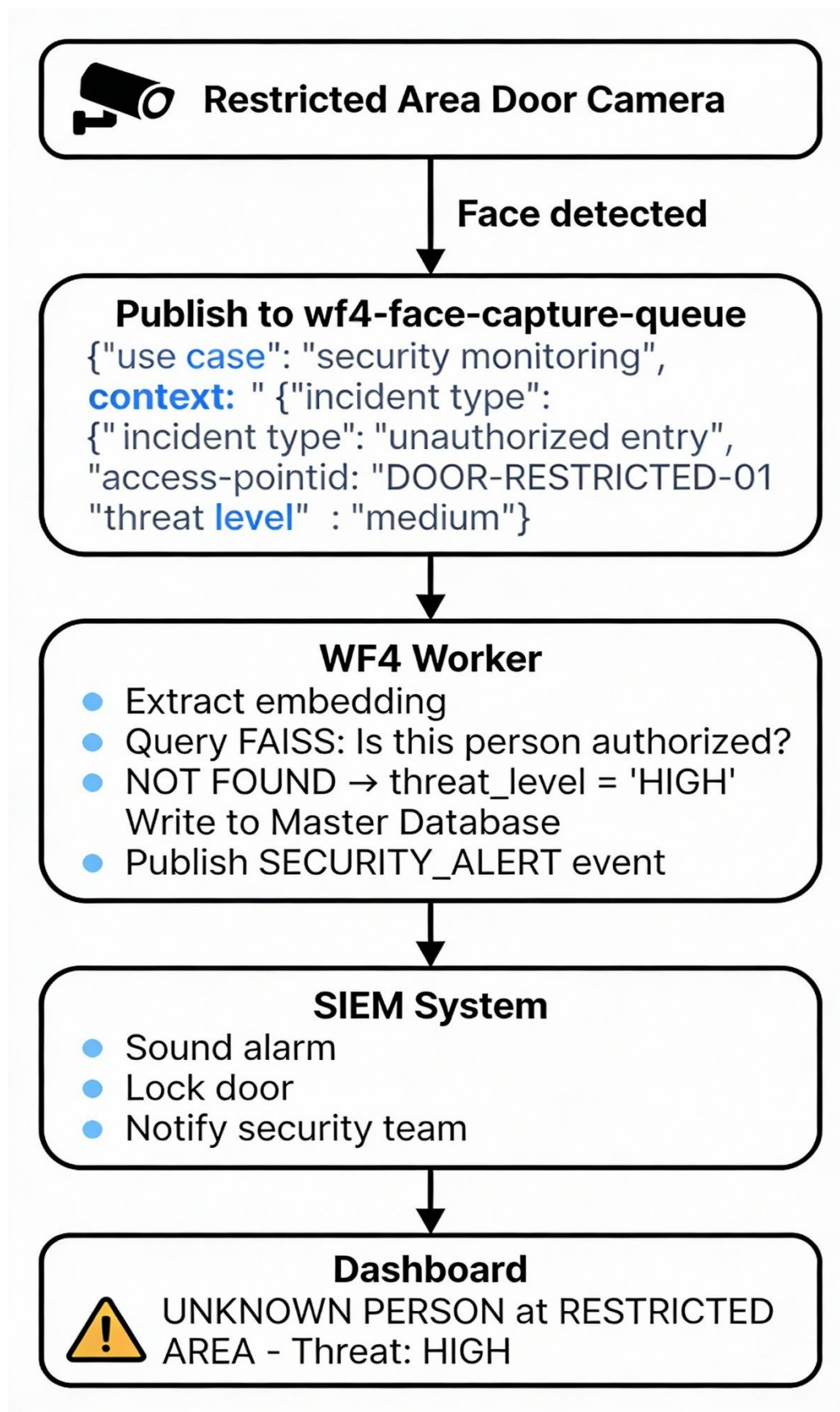
Output

- **Master Database:** Immediate face record with authorization_status, use_case, metadata
- **Persistent FAISS DB:** Embedding available for future searches
- **Event Stream:** Notifications to HR, security, access control, payroll systems
- **Real-Time Dashboards:** Updated statistics and alerts

Use Case: Employee Entry Tracking



Use Case: Security Alert - Unknown Person



Key Metrics

Metric	Value
End-to-End Latency	<1 second
Embedding Extraction	<200ms
FAISS Match	<200ms
Database Write	<100ms
Event Publishing	<50ms
Throughput	10-50 faces/minute (typical), 100+ at peak
Error Rate	<2%
Availability	99.5%
Queue Isolation	Complete - independent of WF1

Configuration :

Workflow 4 – Real-Time Queue & Matching Configuration

Queue Topic:

wf4-face-capture-queue

Queue Retention:

1 hour (shorter than WF1)

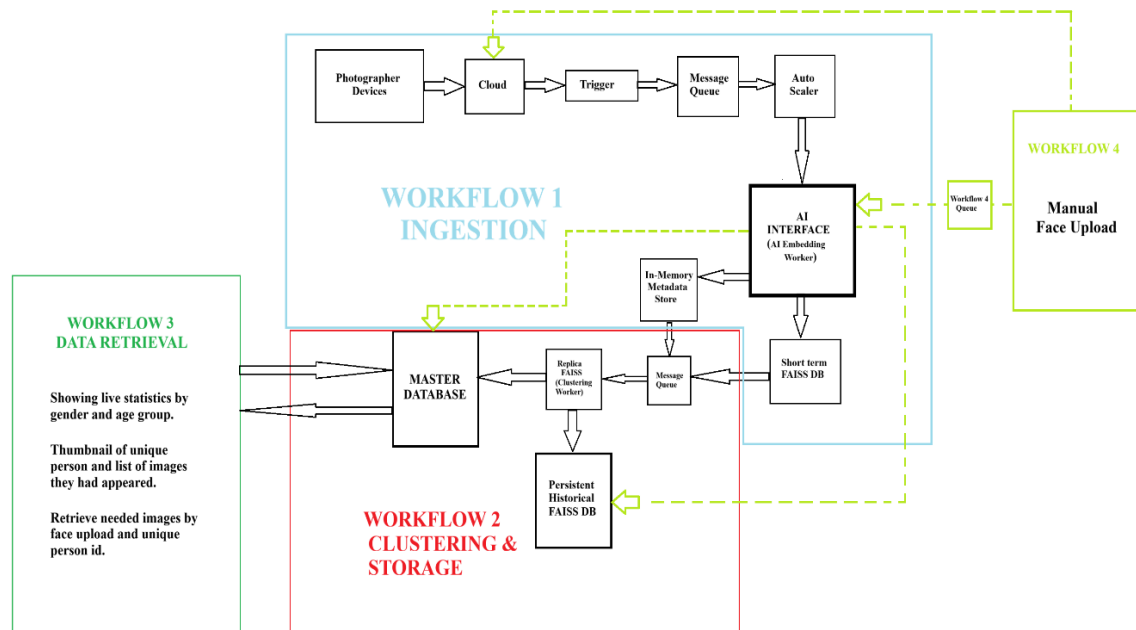
FAISS Threshold:

0.90 (match confidence)

Latency SLA: 1000ms

Max Retry: 3 attempts

ARCHITECTURE DIAGRAM



The diagram shows:

- **WORKFLOW 1 : Photographer Devices → Cloud Storage → Trigger → Message Queue → Auto Scaler → AI INTERFACE**
 - **Output:** Short-term FAISS DB + In-Memory Metadata Store
 - **Feeds:** BATCH_COMPLETED event to WF2
- **WORKFLOW 2 : Replica FAISS (Clustering Worker) ↔ Master Database**
 - **Input:** Message Queue from WF1
 - **Output:** Persistent Historical FAISS DB + Master Database updates
- **WORKFLOW 3 : Data Retrieval reads from Master Database**
 - **Provides:** Live statistics, person galleries, dashboards
- **WORKFLOW 4 : Manual Face Upload (completely separate)**
 - **Independent Queue:** wf4-face-capture-queue
 - **Direct path to:** Master Database + Persistent FAISS DB (no clustering)
 - **Publishes:** Events to external systems (HR, security, access control)

COMPARISON & KEY METRICS

Workflow Comparison Matrix

Aspect	WF1	WF2	WF3	WF4
Type	Batch Ingestion	Deduplication	Analytics	Real-Time Manual
Trigger	Cloud upload	BATCH_COMPLETED	Queries	User action
Entry	Cloud Storage	Message Queue	API	Mobile/Web/Camera
Latency (p95)	500ms-2s/batch	<5s/batch	<500ms	<1s
Throughput	50-200 img/min	1-5 batches/min	100 QPS	10-50 faces/min
Clustering	Yes (DBSCAN)	Core operation	No	No
Queue	wf1-image-batch-queue	Event-driven	N/A	wf4-face-capture-queue
Metadata	Basic	Statistical	Query-based	Rich & use-case specific
External Integration	None	None	None	Yes (6+ systems)
Isolation	Independent	Independent	Read-only	Completely independent
Use Case	Events/Weddings	Uniqueness	Dashboards	Enterprise (6 use cases)

Performance Summary

Metric	WF1	WF2	WF3	WF4
Latency (p95)	800ms	4s	400ms	900ms
Throughput	100-300 img/min	3-10 batches/min	50-200 QPS	10-100 faces/min
Availability	99.5%	99.5%	99.9%	99.5%
Error Rate	<1%	<0.5%	<0.1%	<2%

Metric	WF1	WF2	WF3	WF4
Data Retention	2+ years	Permanent	Read replica	Permanent

Reliability & Isolation

- ✓ **Complete Isolation:** Failure in WF4 doesn't affect WF1, WF2, or WF3
- ✓ **Independent Scaling:** Each workflow scales separately based on demand
- ✓ **Message Retention:** WF1 queue retains 7 days, WF4 retains 1 hour
- ✓ **Dead-Letter Queues:** Failed messages route to DLQ for analysis
- ✓ **Database Replication:** Master DB replicated across regions
- ✓ **Backup Strategy:** Daily snapshots, 30-day retention

DEPLOYMENT & OPERATIONS

Infrastructure Requirements

Compute:

- **WF1:** 2-20 GPU pods (A100, 8GB RAM, 4 vCPU each)
- **WF2:** 1-5 CPU pods (8 vCPU, 16GB RAM each)
- **WF3:** 2-10 API pods (4 vCPU, 4GB RAM each)
- **WF4:** 2-10 GPU pods (A100, 8GB RAM, 4 vCPU each)
- **Total GPU Budget:** 40-80 A100s at peak

Storage:

- **Short-term FAISS:** 50 GB (SSD)
- **Persistent FAISS:** 500 GB - 10 TB (SSD)
- **Master Database:** 100 GB - 1 TB (persistent)
- **Image Cache:** 100 GB - 10 TB (Cloud Storage)
- **Message Queue:** Managed (Pub/Sub)

Networking:

- **Intra-cluster:** 10 Gbps minimum
- **External:** 10 Gbps for image download/upload
- **Database replication:** 1 Gbps minimum

Deployment Checklist

Phase 1: Core Infrastructure

- **Kubernetes cluster (GKE, 40-80 A100 GPUs)**
- **Google Cloud Storage buckets**
- **Pub/Sub topics (wf1, wf4, dead-letter)**
- **PostgreSQL Master Database + replicas**
- **Redis for FAISS write locks**

Phase 2: Workflow Deployments

- **WF1 embedding worker (2-20 replicas, HPA configured)**
- **WF2 clustering worker (1-5 replicas)**
- **WF3 API server (2-10 replicas)**
- **WF4 embedding worker (2-10 replicas)**

Phase 3: Data & Indexing

- **Master Database schema created**
- **Short-term FAISS index initialized**
- **Persistent FAISS index created**
- **Backup policies configured**
- **Data retention policies set**

Phase 4: Monitoring & Ops

- **CloudWatch dashboards per workflow**
- **Alert rules for SLA violations**
- **CloudLogging namespaces configured**
- **Health check endpoints verified**
- **Incident runbooks documented**

Phase 5: Security

- **TLS 1.3 certificates installed**
- **CMEK encryption keys created**
- **IAM roles and service accounts configured**
- **API authentication enabled**
- **Audit logging activated**

Monitoring Metrics

Critical Metrics per Workflow:

- **WF1:** queue_depth, embedding_latency, faces_processed_total, error_rate, gpu_utilization
- **WF2:** clustering_latency, cluster_quality, faiss_index_size, duplicate_rate
- **WF3:** query_latency, dashboard_uptime, data_staleness
- **WF4:** queue_depth, face_latency, match_success_rate, event_publishing_latency

Alert Thresholds:

- **WF1 queue depth > 1000 → Warning (scale up)**
 - **WF4 latency p95 > 1500ms → Critical (investigate)**
 - **Master DB replication lag > 60s → Warning**
 - **FAISS write lock contention > 5s → Critical**
 - **Any error rate > threshold → Page on-call**
-

CONCLUSION

This four-workflow architecture provides a production-grade, enterprise-scale face recognition system that seamlessly handles:

- ✓ **Automatic Batch Processing (WF1):** 1000s of images/minute from multiple photographers
- ✓ **Intelligent Deduplication (WF2):** DBSCAN clustering identifies unique persons
- ✓ **Real-Time Analytics (WF3):** Live dashboards with gender/age/person statistics
- ✓ **Enterprise Use Cases (WF4):** Employee tracking, security, visitors, access control, attendance, events

Key Strengths:

- **Decoupled workflows prevent bottlenecks**
- **Real-time WF4 enables interactive applications**
- **Fault isolation ensures reliability**
- **Elastic scaling matches demand**
- **Event-driven architecture enables external integrations**

Workflow 4 is the game-changer, transforming the system from event-focused to enterprise-ready:

- **Real-time (<1s) manual face uploads**
- **Use-case-aware routing to external systems**
- **Immediate persistence and searchability**
- **Complete isolation from batch processing**

This system is ready for production deployment and scales to handle millions of unique faces across multiple use cases simultaneously.